

6502 Machine Language Opcodes (Alphabetical Order – in hexadecimal)

Opcode	Definition	Immediate	Accumulator	Zero Page	Zero Page,X	Zero Page,Y	Absolute	Absolute,X	Absolute,Y	Indirect Abs.	(Zero Page,X)	(Zero Page),Y	Implied	Relative	Affects Flags
	Abbreviations	imm	acc	zp	zp,x	zp,y	abs	abs,x	abs,y	ind	(zp,x)	(zp),y	imp	rel	
ADC	Add with Carry	69		65	75		6D	7D	79		61	71			N,V,Z,C
AND	Bitwise AND with Accumulator	29		25	35		2D	3D	39		21	31			N,Z
ASL	Arithmetic Shift Left		0A	06	16		0E	1E							N,Z,C
BCC	Branch on Carry Clear													90	
BCS	Branch on Carry Set													B0	
BEQ	Branch on Equal													F0	
BIT	Test Bits			24			2C								N,V,Z
BMI	Branch on Minus													30	
BNE	Branch on Not Equal													D0	
BPL	Branch on Plus													10	
BRK	Break												00		B
BVC	Branch on Overflow Clear													50	
BVS	Branch on Overflow Set													70	
CLC	Clear Carry												18		C
CLD	Clear Decimal												D8		D
CLI	Clear Interrupt												58		I
CLV	Clear Overflow												B8		V
CMP	Compare Accumulator	C9		C5	D5		CD	DD	D9		C1	D1			N,Z,C
CPX	Compare X Register	E0		E4			EC								N,Z,C
CPY	Compare Y Register	C0		C4			CC								N,Z,C
DEC	Decrement Memory			C6	D6		CE	DE							N,Z
DEX	Decrement X												CA		N,Z
DEY	Decrement Y												88		N,Z
EOR	Bitwise Exclusive OR	49		45	55		4D	5D	59		41	51			N,Z
INC	Increment Memory			E6	F6		EE	FE							N,Z
INX	Increment X												E8		N,Z
INY	Increment Y												C8		N,Z
JMP	Jump						4C			6C					
JSR	Jump to Subroutine						20								
LDA	Load Accumulator	A9		A5	B5		AD	BD	B9		A1	B1			N,Z
LDX	Load X Register	A2		A6		B6	AE		BE						N,Z
LDY	Load Y Register	A0		A4	B4		AC	BC							N,Z
LSR	Logical Shift Right		4A	46	56		4E	5E							N,Z,C
NOP	No Operation												EA		
ORA	Bitwise OR with Accumulator	09		05	15		0D	1D	19		01	11			N,Z
PHA	Push Accumulator to Stack												48		
PHP	Push Processor Status to Stack												08		
PLA	Pull Accumulator off Stack												68		N,Z
PLP	Pull Processor Status off Stack												28		ALL
ROL	Rotate Left		2A	26	36		2E	3E							N,Z,C
ROR	Rotate Right		6A	66	76		6E	7E							N,Z,C
RTI	Return from Interrupt												40		ALL
RTS	Return from Subroutine												60		
SBC	Subtract with Carry	E9		E5	F5		ED	FD	F9		E1	F1			N,V,Z,C
SEC	Set Carry												38		C
SED	Set Decimal												F8		D
SEI	Set Interrupt												78		I
STA	Store Accumulator			85	95		8D	9D	99		81	91			
STX	Store X Register			86		96	8E								
STY	Store Y Register			84	94		8C								
TAX	Transfer A to X												AA		N,Z
TAY	Transfer A to Y												A8		N,Z
TSX	Transfer Stack Pointer to X												BA		
TXA	Transfer X to A												8A		N,Z
TXS	Transfer X to Stack Pointer												9A		
TYA	Transfer Y to A												98		N,Z

The result of the opcode directly modifies....

System Memory (Gray 2)
8-bit Registers (Accumulator/X/Y) (Sky blue 10)
16-bit Registers (AB, CD, EF, etc.) (Magenta 10)
CPU Status Register (Orange 1)
Stack/Stack Pointer (Red 2)
Program Counter (Green 2)

Flags

N	Negative (1 = negative)
V	Overflow (1 = true)
B	BRK Instruction (1 = BRK, 0 = IRQB)
D	Decimal Mode (1 = true)
I	IRQB Disable (1 = disable)
Z	Zero Result (1 = true)
C	Carry (1 = true)

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This table originally authored by Lawrence Woodman (https://github.com/lawrencewoodman/machine_language_aids)

Opcode color coding and formatting changes by Brian Manning (https://github.com/spicyjack/8-16_bit_cpu_info)

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6502 Machine Language Opcodes (By Opcode Group - in hexadecimal)

Opcode	Definition	Immediate	Accumulator	Zero Page	Zero Page,X	Zero Page,Y	Absolute	Absolute,X	Absolute,Y	Indirect Abs.	(Zero Page,X)	(Zero Page),Y	Implied	Relative	Affects Flags
	Abbreviations	imm	acc	zp	zp,x	zp,y	abs	abs,x	abs,y	ind	(zp,x)	(zp),y	imp	rel	
Load/Store/Transfer															
LDA	Load Accumulator	A9		A5	B5		AD	BD	B9		A1	B1			N,Z
LDX	Load X Register	A2		A6		B6	AE		BE						N,Z
LDY	Load Y Register	A0		A4	B4		AC	BC							N,Z
STA	Store Accumulator			85	95		8D	9D	99		81	91			
STX	Store X Register			86		96	8E								
STY	Store Y Register			84	94		8C								
TAX	Transfer A to X												AA		N,Z
TAY	Transfer A to Y												A8		N,Z
TSX	Transfer Stack Pointer to X												BA		
TXA	Transfer X to A												8A		N,Z
TXS	Transfer X to Stack Pointer												9A		
TYA	Transfer Y to A												98		N,Z
Arithmetic/Logic															
ADC	Add with Carry	69		65	75		6D	7D	79		61	71			N,V,Z,C
CMP	Compare Accumulator	C9		C5	D5		CD	DD	D9		C1	D1			N,Z,C
CPX	Compare X Register	E0		E4			EC								N,Z,C
CPY	Compare Y Register	C0		C4			CC								N,Z,C
SBC	Subtract with Carry	E9		E5	F5		ED	FD	F9		E1	F1			N,V,Z,C
Bit Manipulation/Rotate/Shift															
AND	Bitwise AND with Accumulator	29		25	35		2D	3D	39		21	31			N,Z
ASL	Arithmetic Shift Left		0A	06	16		0E	1E							N,Z,C
BIT	Test Bits			24			2C								N,V,Z
EOR	Bitwise Exclusive OR	49		45	55		4D	5D	59		41	51			N,Z
LSR	Logical Shift Right		4A	46	56		4E	5E							N,Z,C
ORA	Bitwise OR with Accumulator	09		05	15		0D	1D	19		01	11			N,Z
ROL	Rotate Left		2A	26	36		2E	3E							N,Z,C
ROR	Rotate Right		6A	66	76		6E	7E							N,Z,C
Increment/Decrement/Loops/Indexes/Offsets															
DEC	Decrement Memory			C6	D6		CE	DE							N,Z
DEX	Decrement X												CA		N,Z
DEY	Decrement Y												88		N,Z
INC	Increment Memory			E6	F6		EE	FE							N,Z
INX	Increment X												E8		N,Z
INY	Increment Y												C8		N,Z
Jumping/Branching/Returns															
BCC	Branch on Carry Clear													90	
BCS	Branch on Carry Set													B0	
BEQ	Branch on Equal													F0	
BMI	Branch on Minus													30	
BNE	Branch on Not Equal													D0	
BPL	Branch on Plus													10	
BVC	Branch on Overflow Clear													50	
BVS	Branch on Overflow Set													70	
JMP	Jump						4C			6C					
JSR	Jump to Subroutine						20								
RTI	Return from Interrupt												40		ALL
RTS	Return from Subroutine												60		
Stack Ops															
PHA	Push Accumulator to Stack												48		
PHP	Push Processor Status to Stack												08		
PLA	Pull Accumulator off Stack												68		N,Z
PLP	Pull Processor Status off Stack												28		ALL
CPU Control (Break, NOP, Set/Clear Flags)															
BRK	Break												00		B
CLC	Clear Carry												18		C
CLD	Clear Decimal												D8		D
CLI	Clear Interrupt												58		I
CLV	Clear Overflow												B8		V
NOP	No Operation												EA		
SEC	Set Carry												38		C
SED	Set Decimal												F8		D
SEI	Set Interrupt												78		I

The result of the opcode directly modifies....

System Memory (Gray 2)

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CPU Status Register (Orange 1)

Stack/Stack Pointer (Red 2)

Program Counter (Green 2)

Flags

N Negative (1 = negative)

V Overflow (1 = true)

B BRK Instruction (1 = BRK, 0 = IRQB)

D Decimal Mode (1 = true)

I IRQB Disable (1 = disable)

Z Zero Result (1 = true)

C Carry (1 = true)

Hex	Instruction	ASCII	Dec	Oct	BinH	BinL	Hex	Instruction	ASCII	Dec	Oct	BinH	BinL
00	BRK	NUL	0	000	0000	0000	40	RTI		64	100	0100	0000
01	ORA (zp,x)	SOH	1	001	0000	0001	41	EOR (zp,x)		65	101	0100	0001
02	-	STX	2	002	0000	0010	42	-		66	102	0100	0010
03	-	ETX	3	003	0000	0011	43	-		67	103	0100	0011
04	-	EOT	4	004	0000	0100	44	-		68	104	0100	0100
05	ORA zp	ENQ	5	005	0000	0101	45	EOR zp		69	105	0100	0101
06	ASL zp	ACK	6	006	0000	0110	46	LSR zp		70	106	0100	0110
07	-	BEL	7	007	0000	0111	47	-		71	107	0100	0111
08	PHP	BS	8	010	0000	1000	48	PHA		72	110	0100	1000
09	ORA imm	HT	9	011	0000	1001	49	EOR imm		73	111	0100	1001
0A	ASL acc	LF	10	012	0000	1010	4A	LSR acc		74	112	0100	1010
0B	-	VT	11	013	0000	1011	4B	-		75	113	0100	1011
0C	-	FF	12	014	0000	1100	4C	JMP abs		76	114	0100	1100
0D	ORA abs	CR	13	015	0000	1101	4D	EOR abs		77	115	0100	1101
0E	ASL abs	SO	14	016	0000	1110	4E	LSR abs		78	116	0100	1110
0F	-	SI	15	017	0000	1111	4F	-		79	117	0100	1111
10	BPL	DLE	16	020	0001	0000	50	BVC		80	120	0100	0000
11	ORA (zp),y	DC1	17	021	0001	0001	51	EOR (zp),y		81	121	0101	0001
12	-	DC2	18	022	0001	0010	52	-		82	122	0101	0010
13	-	DC3	19	023	0001	0011	53	-		83	123	0101	0011
14	-	DC4	20	024	0001	0100	54	-		84	124	0101	0100
15	ORA zp,x	NAK	21	025	0001	0101	55	EOR zp,x		85	125	0101	0101
16	ASL zp,x	SYN	22	026	0001	0110	56	LSR zp,x		86	126	0101	0110
17	-	ETB	23	027	0001	0111	57	-		87	127	0101	0111
18	CLC	CAN	24	030	0001	1000	58	CLI		88	130	0101	1000
19	ORA abs,y	EM	25	031	0001	1001	59	EOR abs,y		89	131	0101	1001
1A	-	SUB	26	032	0001	1010	5A	-		90	132	0101	1010
1B	-	ESC	27	033	0001	1011	5B	-		91	133	0101	1011
1C	-	FS	28	034	0001	1100	5C	-		92	134	0101	1100
1D	ORA abs,x	GS	29	035	0001	1101	5D	EOR abs,x		93	135	0101	1101
1E	ASL abs,x	RS	30	036	0001	1110	5E	LSR abs,x		94	136	0101	1110
1F	-	US	31	037	0001	1111	5F	-		95	137	0101	1111
20	JSR (Space)		32	040	0010	0000	60	RTS		96	140	0110	0000
21	AND (zp,x)		33	041	0010	0001	61	ADC (zp,x)		97	141	0110	0001
22	-		34	042	0010	0010	62	-		98	142	0110	0010
23	-		35	043	0010	0011	63	-		99	143	0110	0011
24	BIT zp		36	044	0010	0100	64	-		100	144	0110	0100
25	AND zp		37	045	0010	0101	65	ADC zp		101	145	0110	0101
26	ROL zp		38	046	0010	0110	66	ROR zp		102	146	0110	0110
27	-		39	047	0010	0111	67	-		103	147	0110	0111
28	PLP		40	050	0010	1000	68	PLA		104	150	0110	1000
29	AND imm		41	051	0010	1001	69	ADC imm		105	151	0110	1001
2A	ROL acc		42	052	0010	1010	6A	ROR acc		106	152	0110	1010
2B	-		43	053	0010	1011	6B	-		107	153	0110	1011
2C	BIT abs		44	054	0010	1100	6C	JMP ind		108	154	0110	1100
2D	AND abs		45	055	0010	1101	6D	ADC abs		109	155	0110	1101
2E	ROL abs		46	056	0010	1110	6E	ROR abs		110	156	0110	1110
2F	-		47	057	0010	1111	6F	-		111	157	0110	1111
30	BMI		48	060	0011	0000	70	BVS		112	160	0111	0000
31	AND (zp),y		49	061	0011	0001	71	ADC (zp),y		113	161	0111	0001
32	-		50	062	0011	0010	72	-		114	162	0111	0010
33	-		51	063	0011	0011	73	-		115	163	0111	0011
34	-		52	064	0011	0100	74	-		116	164	0111	0100
35	AND zp,x		53	065	0011	0101	75	ADC zp,x		117	165	0111	0101
36	ROL zp,x		54	066	0011	0110	76	ROR zp,x		118	166	0111	0110
37	-		55	067	0011	0111	77	-		119	167	0111	0111
38	SEC		56	070	0011	1000	78	SEI		120	170	0111	1000
39	AND abs,y		57	071	0011	1001	79	ADC abs,y		121	171	0111	1001
3A	-		58	072	0011	1010	7A	-		122	172	0111	1010
3B	-		59	073	0011	1011	7B	-		123	173	0111	1011
3C	-		60	074	0011	1100	7C	-		124	174	0111	1100
3D	AND abs,x		61	075	0011	1101	7D	ADC abs,x		125	175	0111	1101
3E	ROL abs,x		62	076	0011	1110	7E	ROR abs,x		126	176	0111	1110
3F	-		63	077	0011	1111	7F	-		127	177	0111	1111

Hex	Instruction	ASCII	Dec	Oct	BinH	BinL	Hex	Instruction	ASCII	Dec	Oct	BinH	BinL
80			128	000	1000	0000	120			193	100	1100	0000
81			129	001	1000	0001	121			194	101	1100	0001
82			130	002	1000	0010	122			195	102	1100	0010
83			131	003	1000	0011	123			196	103	1100	0011
84			132	004	1000	0100	124			197	104	1100	0100
85			133	005	1000	0101	125			198	105	1100	0101
86			134	006	1000	0110	126			199	106	1100	0110
87			135	007	1000	0111	127			200	107	1100	0111
88			136	010	1000	1000	128			201	110	1100	1000
89			137	011	1000	1001	129			202	111	1100	1001
8A			138	012	1000	1010	12A			203	112	1100	1010
8B			139	013	1000	1011	12B			204	113	1100	1011
8C			140	014	1000	1100	12C			205	114	1100	1100
8D			141	015	1000	1101	12D			206	115	1100	1101
8E			142	016	1000	1110	12E			207	116	1100	1110
8F			143	017	1000	1111	12F			208	117	1100	1111
90			144	020	1001	0000	130			209	120	1100	0000
91			145	021	1001	0001	131			210	121	1101	0001
92			146	022	1001	0010	132			211	122	1101	0010
93			147	023	1001	0011	133			212	123	1101	0011
94			148	024	1001	0100	134			213	124	1101	0100
95			149	025	1001	0101	135			214	125	1101	0101
96			150	026	1001	0110	136			215	126	1101	0110
97			151	027	1001	0111	137			216	127	1101	0111
98			152	030	1001	1000	138			217	130	1101	1000
99			153	031	1001	1001	139			218	131	1101	1001
9A			154	032	1001	1010	13A			219	132	1101	1010
9B			155	033	1001	1011	13B			220	133	1101	1011
9C			156	034	1001	1100	13C			221	134	1101	1100
9D			157	035	1001	1101	13D			222	135	1101	1101
9E			158	036	1001	1110	13E			223	136	1101	1110
9F			159	037	1001	1111	13F			224	137	1101	1111
100			160	040	1010	0000	140			225	140	1110	0000
101			161	041	1010	0001	141			226	141	1110	0001
102			162	042	1010	0010	142			227	142	1110	0010
103			163	043	1010	0011	143			228	143	1110	0011
104			164	044	1010	0100	144			229	144	1110	0100
105			165	045	1010	0101	145			230	145	1110	0101
106			166	046	1010	0110	146			231	146	1110	0110
107			167	047	1010	0111	147			232	147	1110	0111
108			168	050	1010	1000	148			233	150	1110	1000
109			169	051	1010	1001	149			234	151	1110	1001
10A			170	052	1010	1010	14A			235	152	1110	1010
10B			171	053	1010	1011	14B			236	153	1110	1011
10C			172	054	1010	1100	14C			237	154	1110	1100
10D			174	055	1010	1101	14D			238	155	1110	1101
10E			175	056	1010	1110	14E			239	156	1110	1110
10F			176	057	1010	1111	14F			240	157	1110	1111
110			177	060	1011	0000	150			241	160	1111	0000
111			178	061	1011	0001	151			242	161	1111	0001
112			179	062	1011	0010	152			243	162	1111	0010
113			180	063	1011	0011	153			244	163	1111	0011
114			181	064	1011	0100	154			245	164	1111	0100
115			182	065	1011	0101	155			246	165	1111	0101
116			183	066	1011	0110	156			247	166	1111	0110
117			184	067	1011	0111	157			248	167	1111	0111
118			185	070	1011	1000	158			249	170	1111	1000
119			186	071	1011	1001	159			250	171	1111	1001
11A			187	072	1011	1010	15A			251	172	1111	1010
11B			188	073	1011	1011	15B			252	173	1111	1011
11C			189	074	1011	1100	15C			253	174	1111	1100
11D			190	075	1011	1101	15D			254	175	1111	1101
11E			191	076	1011	1110	15E			255	176	1111	1110
11F			192	077	1011	1111	15F			256	177	1111	1111

		Second Hex Digit															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
First Hex Digit	0x0	BRK															
	0x1																
	0x2																
	0x3																
	0x4																
	0x5																
	0x6																
	0x7																
	0x8																
	0x9																
	0xA																
	0xB																
	0xC																
	0xD																
	0xE																
	0xF																